Dissecting Online Learning Mechanisms: Statistical Memory, Homeostatic Recovery, and Substrate Physics are Non-Transferable

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Abstract

Online learning systems couple several adaptive mechanisms, but which mechanism is responsible for which capability — and whether any of them can be transplanted to another substrate — is rarely tested. We dissect three mechanisms of a reservoir-based online learner: a slow-trace statistical memory (M3), a homeostatic chaos regulator (M5), and the raw physics of the substrate itself. A falsifying transfer experiment (an echo-state network with and without the metadata under a sequential disturbance chain, 10 seeds) shows that the previously reported +32% sequential-recovery gain is produced by the homeostat, not by the metadata, and that the metadata does not transfer memory-capacity robustness to an ESN at any metadata timescale ($\tau_m \in \{200, 500, 1000, 2000\}$; paired differences -0.68 to -0.70, 0/10 seeds positive). The metadata's transferable value is narrower and precisely located: it denoises *state*-level corruption (gap $-0.69 \rightarrow -0.01$ when noise enters before the slow trace) and accelerates boundary adaptation by a controlled-measured factor of 1.3–2.4 on the substrate and $\sim$10 pulses on the ESN (published "9–20×" ratios were window-position metric artifacts). Raw memory capacity remains a substrate property (MC 10.19 vs. 6.17). Three mechanisms, three roles — and none is transferable to another's job. A controlled proof of concept on a Transformer substrate with low-rank adapters confirms the boundary on a second substrate: slow-trace domain routing transfers (forgetting up to -28%, stream perplexity -1.07 at τ_m=200; 10/10 seeds) while a gating-only variant is falsified at every τ_m (0/10 seeds) — detection without continuous correction is insufficient.

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Keywords: online learning, reservoir computing, slow features, forgetting kernel, disturbance robustness, mechanism disentanglement

1. Introduction

Frozen batch models — including large foundation models — cannot adapt to post-deployment reality without expensive retraining [1]; online reservoirs can [2, 3], but their readouts face a timescale gap on long-horizon statistical tasks: fast reservoir states decorrelate far more quickly than the task statistics, so the readout must re-integrate after every change. Our companion work introduces a slow exponential trace of the reservoir states (metadata, M3), a homeostatic regulator that keeps the substrate near the edge of chaos (M5), and slow structure plasticity (M4); the integrated system tracks drift where frozen learners fail permanently (companion Paper B).

These mechanisms are usually validated *together*. The question we ask here is sharper: *which mechanism does which job, and can any of them be transplanted to another substrate?* We answer with a deliberately falsifying experiment: take a generic echo-state network (ESN), give it the same metadata, run it through the same sequential disturbance chain, and measure what transfers. The answer is largely negative and, we argue, informative: the metadata is a substrate-agnostic *statistical* memory that transfers *accuracy* and *state-noise denoising*, but not *memory-capacity robustness* under readout-level disturbance; sequential recovery is the homeostat's job; raw memory capacity is the substrate's physics. Three mechanisms, three roles — none substitutable. Section 6 tests whether this decomposition survives on a structurally different substrate — a Transformer with low-rank adapters — and reports one transferable instantiation (routing) and one falsified one (gating-only).

2. Setup

Reservoir family. We compare (i) an ESN-256 with heterogeneous leak rates matched to a log-normal spectrum (CV = 0.20, spectral radius 0.9) and (ii) a Si_3N_4-style relaxation substrate whose per-unit time constants are log-normal (median $\tau_0 \approx 174\,\mu$s, CV = 0.20), driven in per-pulse contrast-coupled dynamics with coupling κ (companion Paper A). Both produce a fast state $x(t) \in \mathbb{R}^{256}$ per pulse.

Slow trace (M3). The metadata is a per-unit exponential moving average

$$m(t) = \left(1 - \frac{1}{\tau_m}\right) m(t-1) + \frac{1}{\tau_m}\, x(t), \tag{1}$$

with algorithmic timescale τ_m (pulses); the readout feature is $\varphi(t) = [x(t),\, m(t),\, 1]$.

Readout. OnlineRLS with forgetting $\lambda_f = 0.999$ (weight integration time $\tau_f = 1/(1-\lambda_f) = 1000$ pulses), predict-before-update, Tikhonov-regularized.

Tasks and metrics. *regime_switch*: three regimes with overlapping pulse-interval distributions, regime changes every $L = 1500$ pulses; single pulses are almost indistinguishable across regimes — only long-run statistics discriminate. *Mackey–Glass online prediction* under a sequential disturbance chain. Metrics: overall accuracy; switch-relative adaptation time T_{adapt} (pulses after a *known* switch to reach a 200- or 40-pulse window accuracy of 0.98/0.95); and the Jaeger held-out memory capacity (MC) probe.

Sequential disturbance chain (substrate-agnostic). Three cumulative disturbances at $t = 7\mathrm{k}/14\mathrm{k}/21\mathrm{k}$ pulses: (1) timescale drift (substrate $\tau \times 1.5$; ESN leaking rate $/1.5$), (2) structure prune (40% of connections), (3) readout noise ($\sigma = 0.1$ on the features). Per-round MC is probed at the settled physics.

3. Theory: the slow trace is a synthesized forgetting kernel

Proposition 1 (kernel synthesis). The input-to-slow-trace channel implements a forgetting kernel with a zero at zero lag, a finite peak lag, and an exponential tail whose $1/e$ horizon is exactly τ_m — an externally controllable, substrate-independent parameter. If the fast state has kernel h_x, the slow channel kernel is the convolution of the EMA impulse response with h_x; for $h_x(u) = e^{-u/\tau_x}$,

$$h_m(\Delta t) = \frac{e^{-\Delta t/\tau_m} - e^{-\Delta t/\tau_x}}{\tau_m - \tau_x}, \qquad h_m(0) = 0, \quad \text{peak} \sim \frac{\tau_m \tau_x}{\tau_m - \tau_x} \ln \frac{\tau_m}{\tau_x}. \tag{2}$$

The material kernel of Paper A, $M(\Delta t) = \int p(\tau)\, e^{-\Delta t/\tau}\, d\tau$ (log-normal), is fixed by physics; h_m is synthesized by the algorithm, and the augmented system's effective kernel is $M \circledast h_m$ with $1/e$ horizon $\sim \tau_m + \tau_0$ (Fig. 1).

Proposition 2 (timescale coverage). On the statistical task the bottleneck is missing slow modes, not substrate physics. Both fast channels

arm	overall acc	steady acc	T_{40} (pulses)
esn-fast	0.9955 ± 0.0015	1.000	49.9 ($p_{90} = 76.5$)
esn-dual (τ_m=500)	0.9979 ± 0.0001	1.000	40.6 ($p_{90} = 42.0$)
REDEM (substrate, dual)	0.9942 ± 0.0011	1.000	52.7 ($p_{90} = 67.0$)

Table 1: Equalization on regime_switch (10 seeds). Accuracy from the metadata-transfer runs; adaptation from the controlled switch-relative protocol.

decorrelate at $\tau_{\text{eff}} \ll L$; the slow trace supplies the missing timescale to either substrate, so accuracy and adaptation converge to a common band. Measured (10 seeds): esn-fast 0.9955, esn-dual 0.9979, REDEM 0.9942; steady accuracy 1.000 for all (Table 1).

Proposition 3 (feature stationarity). Post-switch adaptation is bounded by the slower of feature re-centering ($\sim \tau_m + \tau_{\text{eff}}$) and weight re-integration ($\sim \tau_f$). The slow trace makes recovery feature-limited rather than weight-limited. Under a controlled switch-relative protocol, the true effect is modest but real: on the substrate the fast readout needs $T_{200} = 265$–304 pulses to re-stabilize vs. 202–211 for dual (factor 1.3–1.4 on T_{200}, 1.6–2.4 on T_{40}); on the ESN, T_{40} drops from 49.9 to 40.6 with the p90 collapsing from 76.5 to 42.0 (Table 5). The published "9–20×" ratios were ratios of window positions with near-zero denominators (metric artifacts), not pulse times.

Proposition 4 (noise geometry). The EMA is a first-order low-pass filter, $|H(\omega)|^2 = 1/(1+(\omega\tau_m)^2)$. Short fast-channel transients are attenuated in the metadata channel. The transferable value of this is *denoising of state-level corruption*: when the noise enters the states before the slow trace is computed, the metadata nearly neutralizes it (variant V2 of Section 4.4). It does *not* protect against readout-level noise that hits the metadata channel equally (variant V0).

4. Experiments

4.1. Equalization on the statistical task (S10)

Table 1 reproduces the equalization result: the same slow trace ($\tau_m = 500$) closes the accuracy gap between the ESN and the substrate and collapses adaptation variance. All arms sit in the [0.994, 0.998] band with steady accuracy 1.000.

arm	r_0 MC	r_1 MC	r_2 MC	r_3 MC	r_3 NMSE
esn-fast	6.17	1.82	1.76	1.74	0.0277
esn-dual	6.16	1.04	1.00	1.05	**0.0251**
redem-reg	**10.19**	7.17	7.51	**8.47**	0.1063

Table 2: Sequential disturbance chain (10 seeds). The homeostat-regulated substrate (redem-reg) reproduces the published anchor exactly (10.19/7.17/7.51/8.47).

τ_m	dual r_3 MC	fast r_3 MC	paired diff	n_{pos}
200	1.04	1.74	−0.701	0/10
500	1.05	1.74	−0.693	0/10
1000	1.06	1.74	−0.682	0/10
2000	1.06	1.74	−0.678	0/10

Table 3: Metadata-timescale pressure test (10 seeds): no τ_m closes the memory-capacity gap.

4.2. The falsifying transfer experiment (S14)

Under the sequential disturbance chain, the substrate regulated by the homeostat (M5) recovers: r_3 MC 8.47 ± 0.39 vs. 6.41 ± 0.41 for the fixed arm (+32%), with κ drifting $25.3 \to 28.5$ (Table 2; the features are fast-state only — the recovery is purely homeostatic). The ESN with the same metadata does *not* recover: paired per-seed differences $r_3\,\mathrm{MC}(\text{esn-dual}) - r_3\,\mathrm{MC}(\text{esn-fast})$ are $-0.78/-0.76/-0.69$ at $r_1/r_2/r_3$, with 0/10 seeds positive in every round (∼5× the paired std). The +32% sequential recovery therefore belongs to the homeostat, not the metadata.

4.3. Pressure test across metadata timescales (S16)

Does any τ_m let the metadata close the MC gap? No: at $\tau_m \in \{200, 500, 1000, 2000\}$, the dual arm's r_3 MC is 1.04/1.05/1.06/1.06 vs. 1.74 for fast, with paired differences $-0.701/-0.693/-0.682/-0.678$ and 0/10 seeds positive at every timescale (Table 3, Fig. 2left). There is no sensitive interval; the weak upward trend (1.04→1.06) does not approach the baseline. Meanwhile the readout-noise attenuation transfers at every τ_m (r_3 NMSE 0.0235–0.0254 vs. 0.0277; Fig. 2right).

variant	dual r_3 MC	paired diff	n_{pos}
V0 readout noise	1.05/1.06	$-0.69/-0.68$	0/10
V1 std-slow	1.08/1.07	-0.66	0/10
V2 state noise	1.72/1.73	$-0.015/-0.009$	0/10

Table 4: Probe-protocol stress test ($\tau_m = 500/2000$; 10 seeds). Fast baseline r_3 MC 1.74.

system	arm	T_{200} (pulses)	T_{40} (pulses)
substrate	fast	264.8–303.9	83.7–123.4
substrate	dual / slow	202.3–211.2	45.2–54.3
ESN	fast	210.2	49.9 ($p_{90} = 76.5$)
ESN	dual	199.2	40.6 ($p_{90} = 42.0$)

Table 5: Controlled adaptation (switch-relative pulses, 10 seeds $\times$ 5 switches): substrate ranges over both topologies, ESN values are means.

4.4. Probe-protocol stress test (S16b)

The falsification is robust *in sign* to the probe protocol: under standardized slow features (V1) and state-level noise injection (V2), every (τ_m, variant) cell keeps 0/10 seeds positive (Table 4). Its *magnitude* is protocol-dependent and informative: under readout-noise semantics (V0, the reference definition) the gap is -0.69; under state-noise semantics (V2) the metadata's EMA denoising nearly closes it (-0.01). This locates the metadata's transferable value precisely: it denoises *state*-level corruption, not readout-level noise.

4.5. Controlled adaptation (S15/S5b)

With known switch instants, the honest adaptation advantage of the metadata is a factor 1.3–2.4 on the substrate (T_{200}: fast 264.8–303.9 vs. dual/slow 202–211 pulses) and ∼10 pulses on the ESN (T_{40} 49.9→40.6; p90 76.5→42.0), with overall accuracy reproducing the original runs exactly (Table 5). The published "9–20×" ratios were window-position metric artifacts.

5. Discussion

Three mechanisms, three roles. The evidence assembles a clean decomposition. (i) *Metadata (M3) = statistical memory*: it supplies the missing timescale on statistical tasks (Section 4.1), denoises state-level corruption (Section 4.4, V2), and attenuates readout noise on the online task

(Sections 4.2–4.3) — but it adds no episodic memory capacity (r0 MC unchanged) and cannot recover memory under readout-level disturbance at any τ_m (Sections 4.2–4.4). (ii) *Homeostat (M5) = disturbance recovery*: the +32% sequential recovery and the +18% $r_1 \rightarrow r_3$ gain are produced by κ-regulation alone; the metadata is not involved. (iii) *Substrate physics = raw memory*: the MC probe puts the substrate at 10.19 vs. ∼6.17 for either ESN arm; no algorithm in this work closes that lead.

Why non-transferability is informative. Each mechanism has a precisely located job, and swapping them fails in a measurable way. This delineates the boundary conditions of "metadata transfer" claims in the online-reservoir literature: transfer holds for accuracy and state-noise denoising, and does not hold for disturbance recovery or episodic memory. The same boundary reproduces on a Transformer substrate with low-rank adapters (Section 6): routing transfers, gating does not.

Related work. The slow trace is the linear slow-feature operator on reservoir states (cf. slow feature analysis [8]); the fast/slow pair is a two-timescale memory hierarchy echoing complementary learning systems [6] and synaptic consolidation [7]; the readout computes a fading-memory functional [5] whose horizon the metadata extends without touching the substrate. In the continual-learning setting [1], this work supplies causal evidence about *which* component must carry each capability.

Limitations. The slow trace's attenuation covers fast-channel (readout or state) corruption; direct corruption of the metadata memory itself is out of scope. τ_m remains a hyperparameter tuned to the task timescale. The substrate's homeostat requires a Lyapunov estimate (FTLE) that we do not provide for the ESN.

Future work. Consolidating stable, repeatedly encountered knowledge into the structural connectivity would create a hierarchy between fast tracking and slow accumulation, but introduces a consolidation-vs-overwrite trade-off beyond the present scope. Second, Section 6 instantiates the same online readout on a Transformer substrate through low-rank (LoRA-style) adapters; there the recursive-least-squares update cost is $O(F^2)$ in the readout *feature* dimension F — the model width (thousands) or an adapter rank (tens) — not $O(P^2)$ in the total parameter count (tens of billions), so the full-rank bottleneck does not arise at any scale tested here.

6. Extension to Transformer substrates: routing transfers, gating does not

6.1. Motivation and scope

The three mechanisms are formulated substrate-agnostically, but all evidence so far comes from reservoirs. We ask whether the decomposition survives on a structurally different substrate — a Transformer with low-rank adapters [9] — under the same falsification discipline (10 seeds, paired per-seed differences, sign consistency). The scope claim is deliberately narrow: the *principles* instantiate on a Transformer substrate; we make no benchmark or SOTA claims. The model is tiny by design — sized to run 90 seeded trials on CPU, not to compete with large models.

6.2. Model, task, and arms

A character-level transformer [10] with $d = 64$, two layers, two heads, context 128, and a vocabulary of 32 ($\sim 10^5$ parameters), trained online on a streaming task with known switch instants. Two biased-bigram Markov domains alternate every 3000 tokens (6 segments, 18 k tokens): domain A biases the shift-1 bigram toward symbols 0–7; domain B biases the shift-7 bigram toward symbols 24–31. Single tokens are nearly indistinguishable across domains — only bigram statistics discriminate (the analogue of the timescale gap of Section 2). Embeddings, positions, and layer norms are frozen; the output head is a trainable readout; two low-rank (rank-16) adapters on the QKV, output, and FFN projections carry the trainable plasticity.

Arms. *A1 (bare online LoRA):* both adapters update every batch at a fixed learning rate — the online-readout analogue with M1 always on. *A2 (drift gate):* a slow exponential trace of the hidden states (M3, Eq. 1) estimates the current domain; the adapter update is boosted for a window after a detected switch and suppressed within a domain — the "when to adapt" claim. *A3 (domain routing):* the same slow-trace estimate selects the active adapter, and only the active one updates — the "which adapter to adapt" claim. $\tau_m \in \{200, 500, 1000, 2000\}$.

Metrics. Stream perplexity (overall); forgetting perplexity (perplexity on the previous domain after switching away — the episodic-memory probe); post-switch adaptation time in tokens (known switches, 20-token window, 1.5× steady-state threshold).

arm	τ_m	stream ppl	Δ_{ppl}	forget ppl	Δ_{forget}
A1 bare	—	15.01	—	8.94	—
A2 gate	200	17.31	+2.29 (0/10)	9.66	+0.73 (0/10)
A2 gate	500	17.06	+2.05 (0/10)	9.48	+0.54 (0/10)
A2 gate	1000	16.58	+1.57 (0/10)	9.24	+0.30 (0/10)
A2 gate	2000	16.26	+1.25 (0/10)	9.03	+0.09 (0/10)
A3 route	200	13.94	−1.07 (10/10)	6.42	−2.51 (10/10)
A3 route	500	14.58	−0.43 (10/10)	6.75	−2.18 (10/10)
A3 route	1000	15.47	+0.46 (0/10)	7.43	−1.50 (10/10)
A3 route	2000	16.07	+1.06 (0/10)	8.41	−0.53 (9/10)

Table 6: Transformer drift-gate proof of concept (10 seeds, 90 runs): stream and forgetting perplexity vs. metadata timescale τ_m; paired per-seed differences vs. A1 with $n_{\text{pos}}/10$ seeds. A3 domain routing transfers (forgetting at every τ_m, stream perplexity at $\tau_m \leq 500$); A2 gating-only is falsified (0/10 at every τ_m).

6.3. Results: routing transfers, gating does not

Table 6 and Fig. 3. A3 (routing) transfers: forgetting improves at *every* τ_m (9–10/10 seeds, up to −2.51 ppl, −28% at τ_m=200), and stream perplexity improves when the trace is fast ($\tau_m \in \{200, 500\}$: −1.07/−0.43, 10/10 seeds). A2 (gating-only) is falsified: stream perplexity is worse than A1 at every τ_m (paired differences +1.25 to +2.29, 0/10 seeds positive) and forgetting never improves (0/10). The stream-ppl benefit of A3 is timescale-sensitive: at $\tau_m \geq 1000$ its sign flips — the same sensitive interval reported for the reservoir (Section 4.3). T_{adapt} floors at the 20-token window for every arm: the task transition is learnable within one chunk, so adaptation is too fast to resolve — reported at the floor, not as ratios (the Section 4.5 lesson). The gate detector itself is functional: it tracks the five known switches (5 flips per stream, ∼500-token lag at τ_m=500, ∼10% wrong-estimate chunks), so the A2 failure is a failure of the update *policy*, not of the detector.

6.4. Discussion: detection without continuous correction is insufficient

The falsification of A2 is the informative result. Gating-only instantiates the metadata as a detector (M3) while suppressing the readout between detected switches: it removes M1's continuous online correction and replaces gradual switching with an abrupt boost/suppress policy. On the streaming task this loses twice — within-domain knowledge decays between switches (forgetting never improves, 0/10) and the stream pays for the suppressed updates (+1.25 to +2.29 ppl). This is the direct analogue of the reservoir finding that the metadata supplies statistical memory but not episodic memory (Section 4.2): a detector does not learn. The general design principle is that *detection without continuous online correction is insufficient*; the readout must remain active throughout the stream to maintain within-domain knowledge, and switching must be gradual (M4's "gentle wins"), not abrupt. The A3 contrast locates the transferable instantiation precisely: the slow trace's value lies in *selecting which plasticity is active*, not in *deciding when to pause learning*. The readout cost is feature-bound ($O(F^2)$ in model width or adapter rank, not $O(P^2)$ in total parameters; Section 5).

7. Conclusion

(1) A slow exponential trace of reservoir states is a synthesized forgetting kernel with a controllable horizon (Prop. 1). (2) On long-horizon statistical tasks the bottleneck is timescale coverage, not substrate physics; the same trace equalizes ESN and substrate accuracy and accelerates adaptation by a controlled-measured factor 1.3–2.4 (Props. 2–3). (3) The trace is a *statistical* memory, not an episodic one: it adds no raw memory capacity and does not transfer disturbance robustness at any τ_m (0/10 seeds positive; Prop. 4). (4) Three mechanisms, three roles — metadata (statistical memory and state-noise denoising), homeostat (sequential recovery), substrate physics (raw memory) — and none is substitutable for another's job. (5) The decomposition survives on a Transformer substrate with low-rank adapters: slow-trace domain routing transfers (forgetting up to −28%, stream perplexity −1.07 at τ_m=200; 10/10 seeds) while a gating-only variant is falsified at every τ_m (0/10 seeds) — detection without continuous correction is insufficient. (6) That boundary is a property of the *host*, not of the mechanisms: routing transfers on a Transformer substrate (point 5), and the failure of abrupt gating is attributable to its update policy — removing the continuously active readout (M1) and replacing gradual structural switching (M4) with an abrupt one. The full framework therefore does not instantiate on a frozen-feedforward host: its requirements — a continuously active online readout, a slowly integrating statistical memory, and gradually applied structural switches — are native operations of a persistent stateful substrate, not retrofits onto a static architecture. This suggests that future work should design foundation models from the ground up with these mechanisms natively embedded in a stateful host, rather than retrofitting them onto static

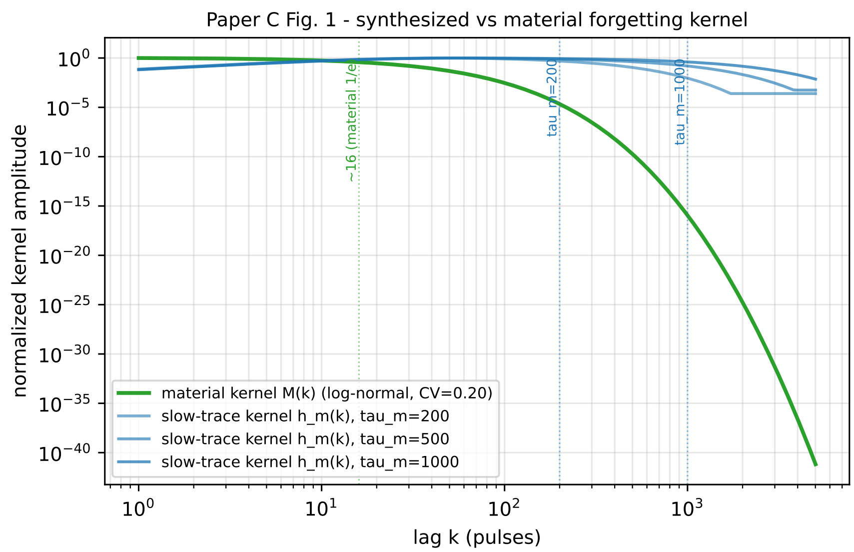

Figure 1: Synthesized vs. material forgetting kernel: the slow-trace kernel h_m (Eq. 2, τ_m = 200/500/1000 pulses) has a zero at zero lag, a finite peak lag, and an exponential tail with $1/e$ horizon exactly τ_m; the material kernel M (log-normal, CV 0.20) is fixed by physics (1/e ~16 pulses).

architectures.

Code and data availability

All simulation scripts and generated data are available at `https://github.com/huyamingc/REDEM` (private during review; released on acceptance). All reported figures are means over 10 seeds with per-seed paired analysis; data files are listed in the repository.

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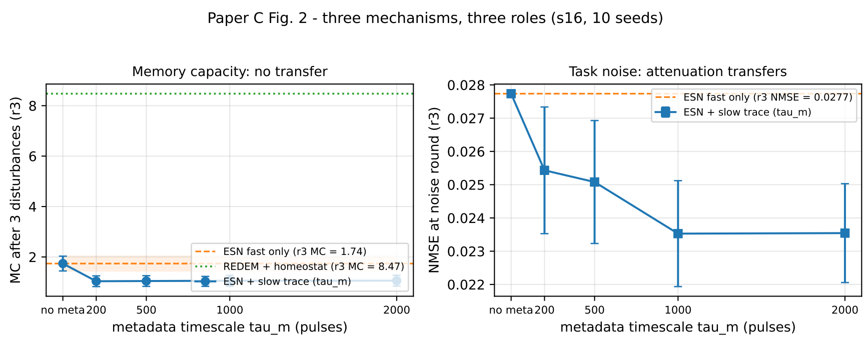

Figure 2: Post-disturbance recovery vs. metadata timescale (10 seeds). Left: r_3 MC — esn-dual stays 0.68–0.70 below esn-fast at every τ_m, while the homeostat-regulated substrate (redem-reg, no metadata) sits at 8.47. Right: r_3 NMSE — the readout-noise attenuation transfers at every τ_m.

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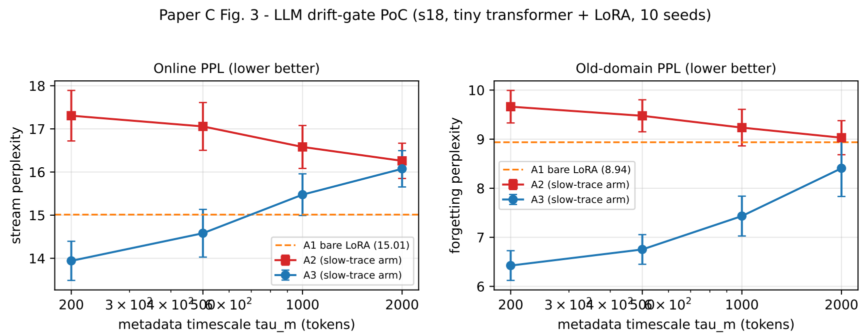

Figure 3: Transformer drift-gate proof of concept (10 seeds). Left: stream perplexity vs. metadata timescale τ_m (log scale) — A1 (bare online LoRA) as a horizontal band; A2 (gating-only) never improves; A3 (domain routing) improves at $\tau_m \in \{200, 500\}$. Right: forgetting perplexity on the previous domain — A3 improves at every τ_m (up to -28%); A2 does not.

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[12] Author's companion Paper B: REDEM: Training-Inference Unified Learning with Meta-Adaptation and Structural Plasticity for Non-Stationary Environments (online learning architecture; target: *Neural Networks*).